Sahaj Bhadja

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Summary

GovTech-oriented software engineer with 2+ years building **Java/Python services, distributed systems, secure APIs, and payment workflows**; experienced in OOP, design patterns, testing, production reliability, documentation, Linux, Git, and AI-assisted development.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **secure microservices and REST APIs** across Python, FastAPI, PostgreSQL, SQL Server, and cloud storage, supporting 500+ users while reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Refactored distributed processing logic into reusable services with explicit states, validation, and failure isolation, improving throughput 3x while sustaining 99%+ availability across 5,000+ monthly production workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **user-facing React interfaces and Flask services** for uploads, processing, review, and feedback, serving 150+ users and 10,000+ daily requests through authenticated REST APIs on AWS.
- Used AI-assisted development with Cursor, OpenAI, and Claude to generate, debug, test, and optimize code, verifying outputs through unit and integration tests, code reviews, and production monitoring.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Developed **Java/Python microservices and third-party integrations** using Singleton and Factory patterns, REST APIs, SQL, caching, AWS Glue, and S3, improving response times 30% and pipeline efficiency 68%.
- Maintained 20+ unit and integration tests with PyTest and Mockito across Linux, Windows, and macOS while collaborating through Git, code reviews, Jenkins automation, and documented architecture decisions.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built **secure client-facing web applications** across React, Node.js, eight microservices, and Azure SQL for authentication, records, billing, permissions, communications, and notifications, securely synchronizing 2M+ records across locations.
- Kept reliable production systems healthy through Docker, Jenkins, automated tests, caching, and observability with Sentry, Azure Monitor, and Kibana, reducing deployment time 50% and resource waste 42%.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Built **secure digital payment and service workflows** for pricing, payments, refunds, disputes, and status reporting, supporting 50,000+ records through Flask APIs, Firebase synchronization, and sub-200-millisecond responses.
- Integrated secure third-party and cross-platform systems through REST APIs, webhooks, and data pipelines, documenting 12+ architecture flows while reducing manual refund handling 70% and improving transaction correctness.

Skills

Languages: Java, Python, JavaScript, TypeScript, SQL, Bash
Core Engineering: OOP, Singleton/Factory patterns, microservices, distributed systems, refactoring, Unix/Linux
Integration: REST APIs, web services, JWT authentication, third-party systems, webhooks, ETL, SQL/NoSQL
Quality: Unit/integration testing, PyTest, Mockito, code reviews, correctness, security, availability, documentation
Delivery: AWS, Azure, Docker, Kubernetes, Git, Jenkins, CI/CD, monitoring, Cursor, Claude, AI-assisted development

Projects

HMIS Live | *Digital Services, APIs, Security*

- Built a **secure digital-service workflow application** with role-based access, dashboards, appointments, REST integrations, notifications, and Sentry monitoring, halving record retrieval time while supporting 100+ daily workflows and reducing administration 30%.

FinNexus Live | *Payments, TypeScript, Kubernetes*

- Developed a **scalable digital payment platform** supporting 50,000+ users and 1M+ transactions with secure bank integrations, token-based access, real-time updates, advanced filtering, MongoDB persistence, and Kubernetes-based automation.

Chat2SQL Code | *Java, TypeScript, System Integration*

- Enhanced a **cross-platform integration application** supporting 10+ SQL and NoSQL systems with schema exploration, compatibility checks, query generation, validation, Docker configurations, and reusable development and deployment workflows.